Digital Twin: Research Summary

2026-10-18. Detailed reports: 01-landscape · 02-hardware · 03-algorithms-software

What a "complete" digital twin is made of

A digital twin has five layers. Most products on the market cover only one or two.

Layer	What it is	Update rate	Main tech
1. Geometry	Metric 3D model of the space (point cloud, mesh)	Weekly or per capture	LiDAR SLAM, photogrammetry, TLS
2. Appearance	Photoreal, walkable view	Weekly	360° imagery, Gaussian splatting
3. Semantics	Labelled objects (walls, ducts, rebar, machines), linked to BIM elements	Per capture	Point-cloud segmentation, SAM 3 / Grounding DINO lifted to 3D, IFC
4. Time (4D)	History of changes compared with the plan: progress, deviation, schedule	Per capture plus history	Scan-to-scan (M3C2), scan-to-BIM, P6/MSP linking
5. Live state	Where workers and equipment are, sensor readings, alerts	Seconds to minutes	Fixed cameras, UWB/BLE, telematics, IoT over MQTT

What others are doing (summary)

- **Most money is in progress tracking against BIM and the schedule**, captured by a person walking with a 360° helmet camera each week. Players: OpenSpace, Buildots (~$1B valuation), Doxel, Track3D and Cupix. Data centers are the leading vertical.
- **Capture is now a commodity.** Track3D gives documentation away free and charges $0.15/sq ft for progress analytics.
- **Autonomous capture is the new frontier, but still early.** DroneDeploy's Capture Agent (Spot/Unitree, navigates from drawings) is in early access. Spot costs $150k+.
- **Live data sits in silos.** Crane cameras (Versatile), safety CV (Intenseye), telematics (Propeller DirtMate) and fixed cameras (OxBlue/Evercam) are not fused into one spatial twin.
- **Incumbents keep buying startups:** Hexagon, Trimble, Bentley+Cesium, Autodesk, AMETEK/FARO, CoStar/Matterport.

Opportunities that fit Flo Mobility (robotics)

1. **Autonomous ground-robot capture sold as a service.** A wheeled AMR (later legged) repeats a capture route nightly without a human. This makes capture cheap and frequent, and turns a weekly twin into a daily one.
2. **A fused live twin.** The robot's periodic scans become the geometry backbone, and cameras, UWB tags and telematics become the live layer on top.

3. **Trustworthy analytics.** Element-level progress and deviation with confidence scores, and occluded elements reported as unknown.
4. **A twin built for robots.** The same map serves navigation and simulation (Isaac Sim / USD), not just human viewing.

Bill of materials: what we need

Sensors (MVP to mid tier on our robot)

Item	Pick	Approx. cost
3D LiDAR	Hesai JT128 (hemispherical) or Livox Mid-360 (cheap, FAST-LIO standard)	$1-1.5k
Cameras	ZED X stereo (GMSL) + Insta360 X5 360° camera	~$1.1k
Positioning	Unicore UM980 / u-blox F9P RTK on the robot + Emlid RS3/RS4 for ground control	$0.3k + $3-4k
Edge compute	Jetson AGX Orin 64 GB (Orin NX for MVP)	$1-3.5k
Storage	2-4 TB NVMe on the robot	~$0.3k
Exterior	DJI Matrice 4E drone (or Skydio, for NDAA compliance)	~$5k
Reconstruction	Workstation with RTX 5090/6000-class GPU and 128 GB+ RAM, or cloud GPUs	$6-15k
Accuracy check	Rented TLS (RTC360) for ground truth	~$500-1k/day

Expected accuracy: **1-2 cm relative, 2-3 cm absolute**. That is enough for progress tracking (LOA10-20). It is not enough for QA against BIM, which needs 5-15 mm (LOA30) and requires TLS or a partnership.

Live layer (phase 2)

Fixed or crane cameras, UWB badges (~$25-99 per tag, $500-800 per anchor), ISO 15143-3 (AEMP) telematics APIs, concrete maturity and environmental sensors over LoRaWAN, and Starlink or 5G backhaul.

Algorithms and software (open-source first)

Stage	Pick
Robot middleware	ROS 2 Jazzy, MCAP, Foxglove/Rerun
Odometry / SLAM	FAST-LIO2 / FAST-LIVO2 → GLIM (global, multi-session, GNSS)
Drone	WebODM
Point clouds	PDAL + COPC, Open3D, CloudCompare, small_gicp, TEASER++
Visual twin	gsplat, seeded with LiDAR points (Apache; avoid Inria's non-commercial code)

Stage	Pick
Dynamic-object removal	MapMOS / DUFOMap (keep removed objects as the safety layer)
Semantics	Pointcept (PTv3/Sonata), SAM 3 + Grounding DINO / RF-DETR
Change and progress	py4dgeo M3C2, IfcOpenShell C2M per IFC GUID, MPXJ for P6/MSP schedules
Twin state and IoT	Eclipse Ditto, MQTT → Kafka/Redpanda → TimescaleDB + PostGIS
Viewer	CesiumJS (3D Tiles + splats), Potree, That Open Engine (IFC)
Formats	E57, LAZ/COPC, IFC 4.3, glTF + KHR_gaussian_splatting/SPZ, 3D Tiles, OpenUSD

Licensing to watch: ORB-SLAM3 is GPL. OpenMVS and Ultralytics YOLO are AGPL. Inria 3DGS is non-commercial. FAST-LIO is GPL-2.0, which matters if we ship it on a product.

Reference pipeline

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Robot/drone/360/TLS capture ─▶ Ingest (MCAP/E57/LAZ → S3) ─▶ Align (LIO →
GLIM merge → RTK/GCP georef → BIM registration)
   ─▶ Reconstruct (COPC, splats, mesh → 3D Tiles) ─▶ Semanticize (segment +
remove dynamics + link to IFC)
   ─▶ Analyze (scan-to-scan, scan-to-BIM, progress → schedule, safety
zones) ─▶ Serve (web viewer, API) ─▶ Alert
Edge: drivers, real-time LIO, PPE/equipment detection, blurring,
compression.   Cloud: everything heavy.
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Proposed phased plan

Phase	Goal	Outcome
0. Rig and data (2-4 wks)	Mid-360/JT128 + cameras + RTK on a Flo robot (or handheld pole); record MCAP	Repeated captures of one test site
1. Geometry twin (4-6 wks)	FAST-LIO2 → GLIM, georeferenced COPC, Potree/Cesium viewer	A measurable 3D site you can view in a browser
2. Visual + 4D (4-6 wks)	gsplat from the same captures; scan-to-scan change maps over time	A photoreal twin with a timeline slider and change heat maps
3. Semantics + BIM (6-8 wks)	IFC registration, per-element progress and deviation, schedule link	Progress reports
4. Autonomy + live layer	Nightly autonomous capture routes; cameras, UWB, telematics; alerts	A near-live twin; robotics-as-a-service pitch

Decisions needed from the team

1. Main user and buyer: GC project managers (progress), safety teams, owners, or our own robots.
2. Update rate: nightly autonomous capture vs. live (cameras + tracking) first.
3. Target site types: building interiors (BIM-heavy) vs. earthworks, solar and civil (underserved).
4. Reuse of existing Flo Mobility robot hardware.